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Yuxuan Lou

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Research Interest

- Diffusion Large Language Models
- Efficient Large Language Model Scaling with Mixture of Experts
- Multimodal Foundation Model Adaptation from Large Language Models

Education

National University of Singapore, School of Computing, HPC-AI Lab — 2023 – Present
- Ph.D. in Computer Science, Advised by Prof. Yang You

National University of Singapore, School of Statistics and Probability — 2020 – 2022
- M.Sc. in Statistics

Harvard University, Computer Science Department, DAS Lab — 2019 – 2020
- Research Intern

Fudan University, School of Mathematical Science — 2016 – 2020
- B.Sc. in Applied Mathematics

Research Experiences

Diffusion Large Language Models, InclusionAI, Ant Group — Apr 2026 – Sep 2026
- Research Intern on diffusion large language models
- Contributed to **LLaDA-Image**, a 6B open-source unified image generation and editing model family built with fully open training recipes (arxiv.org/abs/2609.03796)

Diffusion-based Speech-Text Language Model, NUS - Tencent — Sep 2025 – Mar 2026
- Developed **DiffuSpeech**, the first diffusion-based speech-text language model supporting both understanding and generation, introducing a "Silent Thought, Spoken Answer" paradigm where internal text reasoning informs spoken responses
- Unified discrete text and tokenized speech under a single masked diffusion framework with modality-specific masking schedules, enabling joint generation of reasoning traces and speech tokens through iterative denoising
- Constructed **ThinkingTalk**, the first speech QA dataset with paired text reasoning traces (26K samples, 319 hours), achieving state-of-the-art speech-to-speech QA accuracy (+9 points over best baseline) and best TTS quality among generative models (6.2% WER)

Efficient Foundation Models with Mixture of Experts, NUS - Apple — Sep 2024 – May 2025
- Developed **MoST**, a novel speech-text foundation model featuring a Modality-Aware Mixture of Experts (MAMOE) architecture which directs tokens to specialized pathways for enhanced cross-modal understanding; achieved competitive performance across multiple speech-text benchmarks using exclusively open-source data
- Designed modality-specific expert routing strategies and shared-expert configurations that improve cross-modal knowledge integration while maintaining inference efficiency

Multimodal LLM Agent with Retrieval Augmented Planning, NUS - Panasonic — Oct 2023 - May 2024
- Developed **RAP**, a Multimodal planning agent which leverages past successful experiences to enhance decision-making process
- Developed **EnvBridge**, a Multimodal embodied agent which can transfer knowledge from diverse embodied environments and enhance planning ability
- SOTA results on text-only environments(ALFWorld, Webshop), Significant improvements on multimodal robotics benchmarks(Franka Kitchen, Meta-World, RLBench)

Vision Model Scaling with Mixture of Experts, HPC-AI Lab — Mar 2021 – Jan 2022

- Developed large-scale vision models: **Sparse-MLP**, **Widenet** based on Mixture of Experts
- Proposed a fully-MLP architecture with conditional computation in two directions and extended MoE to spatial dimension of image representation

Selected Publications

*Equal contribution †Corresponding author

DivMoE: Fine-Grained MoE Upcycling via Cross-Domain Expert Composition(NeurIPS 2026)
Yuxuan Lou*, Kai Yang*, Geng Zhang, Yong Liu, Yang You
arxiv.org/abs/2610.11317

OrScale: Orthogonalised Optimization with Layer-Wise Trust-Ratio Scaling(2026)
Yuxuan Lou, Yang You
arxiv.org/abs/2605.07815 · Github

AsyncLane: Decoupling Refinement from Advancement in Diffusion Language Model Decoding(2026)
Yingxuan Ren*, **Yuxuan Lou***, Yong Liu, Pengcheng Fang, Ziming Wang, Pengfei Zhou, Yang You†
arxiv.org/abs/2606.08411

ECUpcycle: Slice-Structured Expert-Choice Diffusion Language Model Upcycling(2026)
Yong Liu*, **Yuxuan Lou***, Hailun Xu, Yingxuan Ren, Haoying Li, Kanchan Sarkar, Kun Xu, Yingyan Celine Lin, Cho-Jui Hsieh, Yang You

DiffuSpeech: Silent Thought, Spoken Answer via Unified Speech-Text Diffusion(2026)
Yuxuan Lou*, Ziming Wu*, Yaochen Wang, Yong Liu, Yingxuan Ren, Fuming Lai, Shaobing Lian, Jie Tang, Yang You
arxiv.org/abs/2601.22889

MoST: Modality-Aware Mixture of Experts for Efficient Speech-Text Foundation Model(ICML 2026)
Yuxuan Lou*, Kai Yang*, Yang You
arxiv.org/abs/2601.10272 · Github

MoNE: Replacing Redundant Experts with Lightweight Novices for Structured Pruning of MoE(ICLR 2026)
Geng Zhang, Yuxuan Han, **Yuxuan Lou**, Yiqi Zhang, Wangbo Zhao, Yang You
arxiv.org/abs/2507.00390

EnvBridge: Bridging Diverse Environments with Cross-Environment Knowledge Transfer for Embodied AI(2024)
Tomoyuki Kagaya*, **Yuxuan Lou***, Thong Jing Yuan*, Subramanian Lakshmi*, Jayashree Karlekar, Sugiri Pranata, Natsuki Murakami, Akira Kinose, Koki Oguri, Felix Wick, Yang You
arxiv.org/abs/2410.16919

RAP: Retrieval-Augmented Planning with Contextual Memory for Multimodal LLM Agents(2024)
Tomoyuki Kagaya*, Thong Jing Yuan*, **Yuxuan Lou***, Jayashree Karlekar*, Sugiri Pranata, Akira Kinose, Koki Oguri, Felix Wick, Yang You
arxiv.org/abs/2402.03610

Cross-token modeling with conditional computation(2022)
Yuxuan Lou, Fuzhao Xue, Zangwei Zheng, Yang You
arxiv.org/abs/2109.02008

Funded Research & Awards

2026 Google Research Award – Project Lead (PI: Prof. Yang You) Apr 2026

- "Pioneering the Systems Foundations for Diffusion-Based Large Language Models on TPUs"; received $100K in Google Cloud credits
- Project lead under PI Prof. Yang You; resulting publications: DiffuSpeech, AsyncLane, ECUpcycle

NUS Research Achievement Award – National University of Singapore Aug 2026

Open Source Projects

Colossal-AI: Making large AI models cheaper, faster, and more accessible 41k star

- A collection of parallel components for distributed training of large deep learning models
- Managed and contributed to Colossal-AI examples

awesome mixture-of-experts 1.2k star

- A collection of awesome Mixture of Experts papers and projects

ClawFinetune

- Portable fine-tuning skill package for agent-driven workflows; generates and launches LoRA SFT runs on the Tinker and HPC-AI SDK backends with preflight validation and run-state tracking

ClawTrade

- Secure multi-broker trading middleware that lets AI agents operate brokerage accounts behind a containerized, guardrail-enforced HTTP API

Invited Talks

Meta MRS Team – *Diffusion LLMs & Multimodal MoE: Pushing the Frontier of Multimodal Foundation Models* Apr 22, 2026

Apple GPX APAC Sharing Session – *Mixture of Experts for Multi-modal* Sep 17, 2025

Skills & Technologies

GPU Training: PyTorch, DeepSpeed, Megatron-LM, Colossal-AI, HuggingFace Transformers/Accelerate, vLLM, FlashAttention (NVIDIA GPU clusters)

TPU Training: TensorFlow, JAX/Flax, Keras (Google Cloud TPU pods)

Parallel Training & Optimization: Model parallel, tensor parallel, pipeline parallel, sequence parallel, data parallel, mixture-of-experts parallel training